EC 571: Digital VLSI Circuit Design

CMOS Power, Wires

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Acknowledgement - Some slides adapted from:

- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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Last lecture

- Last lecture
  - CMOS delay
- Today's lecture
  - CMOS power
  - Wires
- Reading
  - Chapter 4

CMOS Power Dissipation

- Dynamic power
- Short circuit power
- Leakage power
- Static power.

Dynamic energy consumption

$$E_{0 \to 1} = C_L V_{DD}^2$$

$$E_C = \frac{1}{2} C_L V_{DD}^2$$

Dynamic Power Consumption

Power = Energy / transition * Transition rate.

transitions / sec.

$$= C_L V_{DD}^2 f_{0\to1}$$

$$= C_L V_{DD}^2 f P_{0\to1}$$

$P_{0\to1}$: Probability of a $0\to1$ transition

$$= C_{switched} V_{DD}^2 f$$

$C_{switched} = C_L \cdot P_{0\to1}$

Transition Activity & Power

- Energy consumed in 'N' cycles

$$E_N = C_L V_{DD}^2 \, n_{0 \to 1}$$

$n_{0 \to 1}$: Number of $0 \to 1$ transitions in N cycles.

$$P_{avg} = \lim_{N \to \infty} \frac{E_N}{N} \cdot f$$

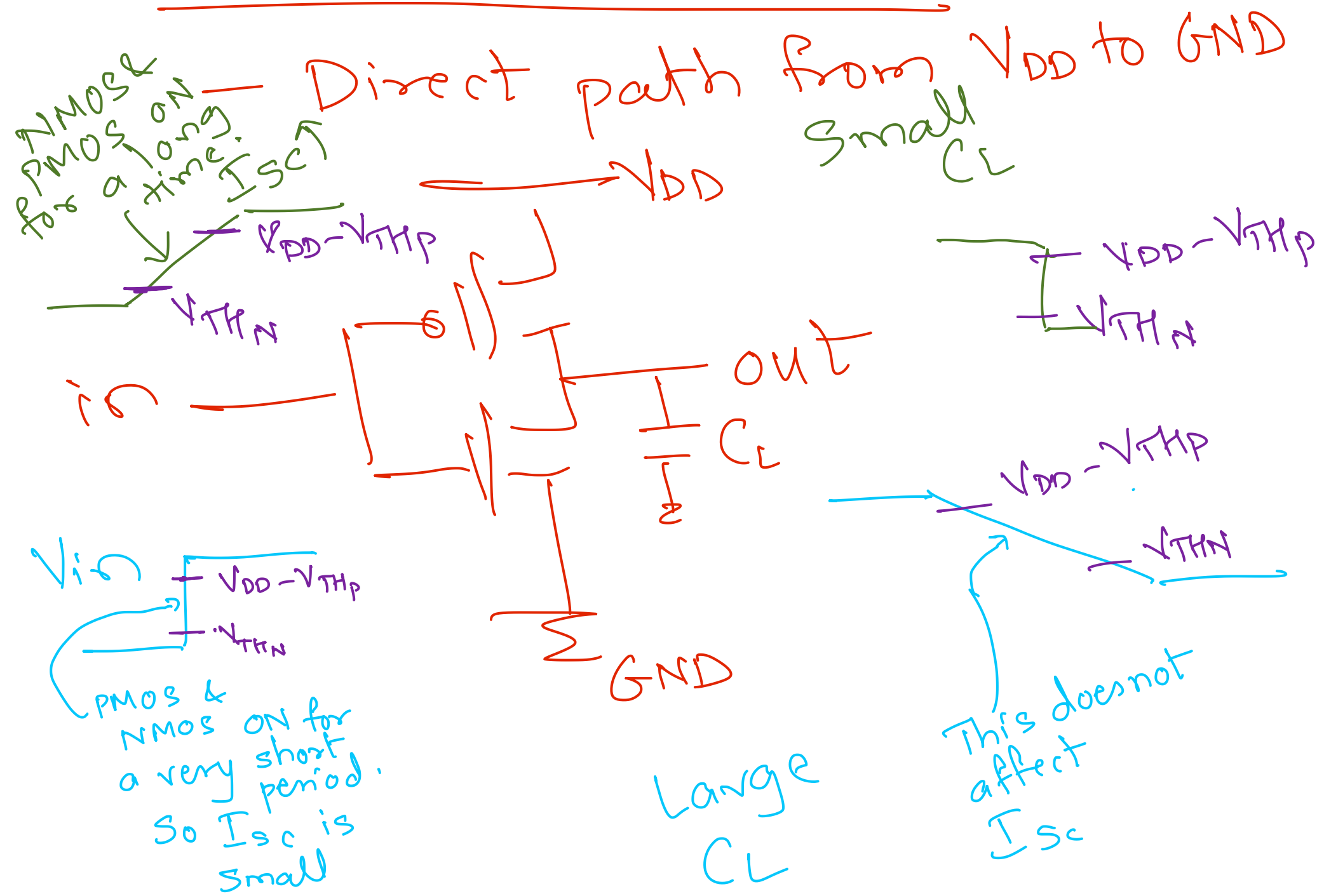
$$= \lim_{N \to \infty} \frac{n_{0 \to 1}}{N} C_L V_{DD}^2 f$$

$$\lim_{N \to \infty} \frac{n_{0 \to 1}}{N} = \alpha_{0 \to 1}$$

activity factor for $0 \to 1$ transitions

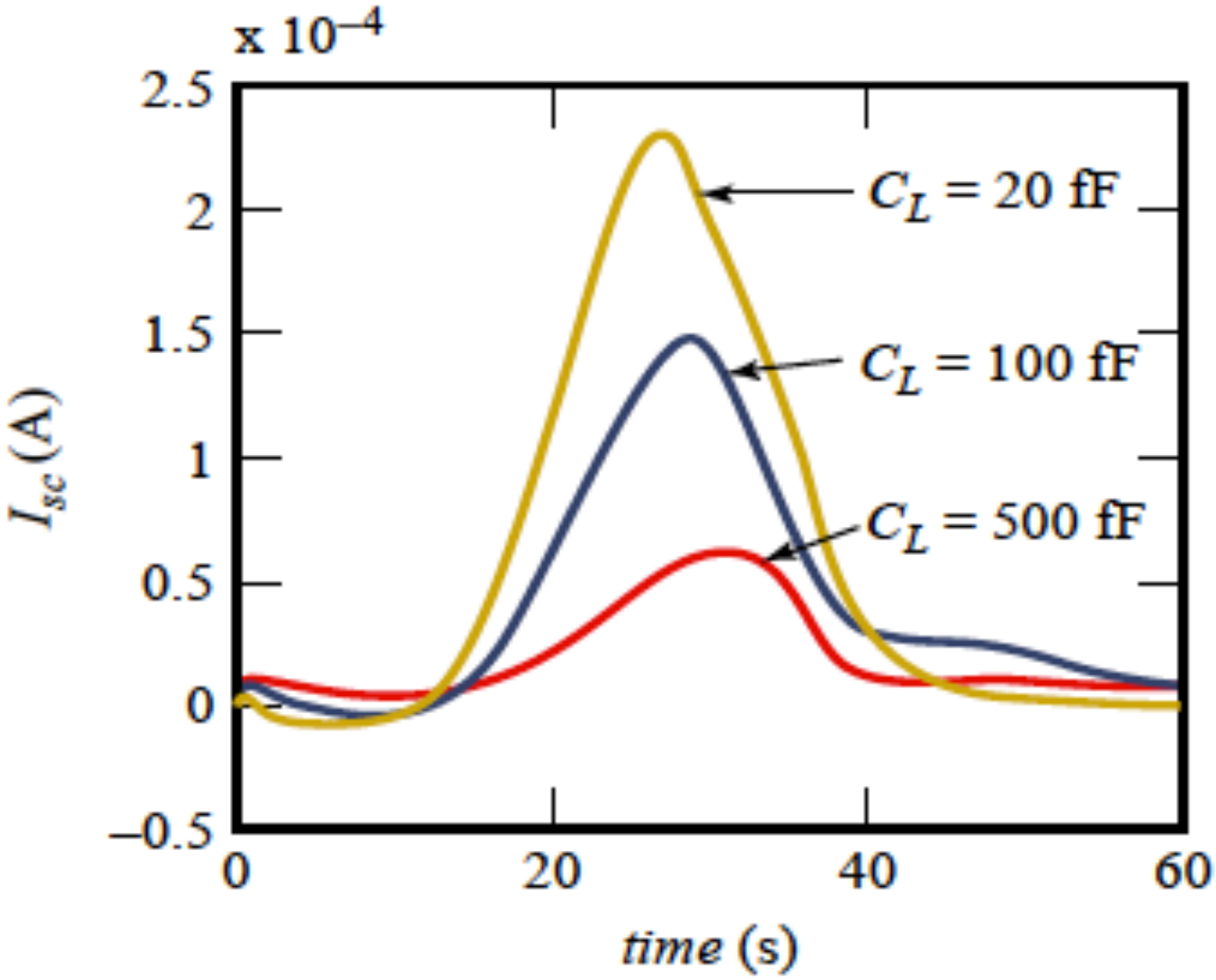
$$P_{avg} = \alpha_{0 \to 1} C_L V_{DD}^2 f$$

Short circuit Power



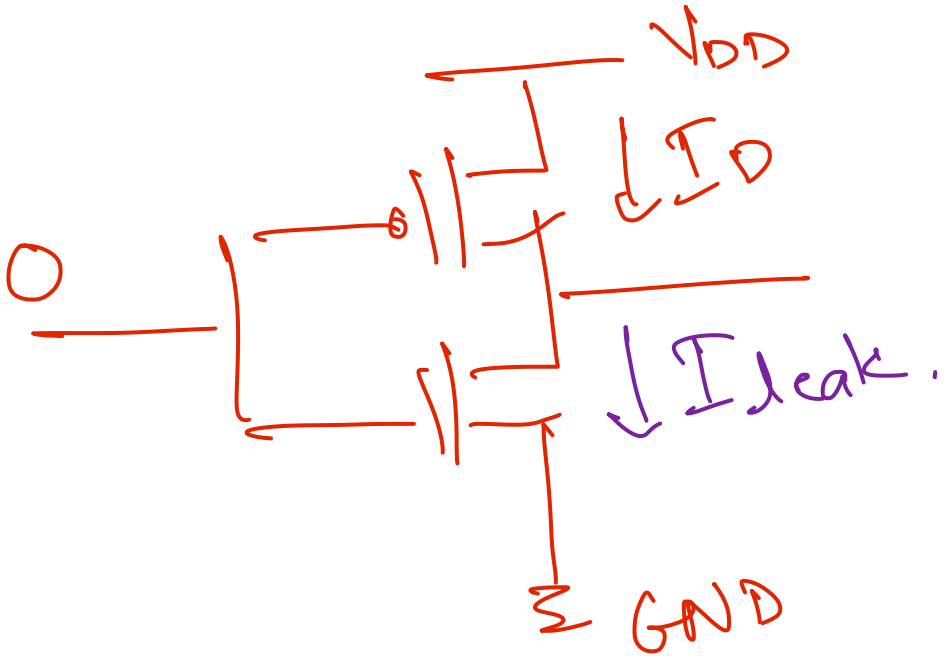
Slew rate / Rise time / Fall time
are important
— Affect Performance
& Short ckt. power

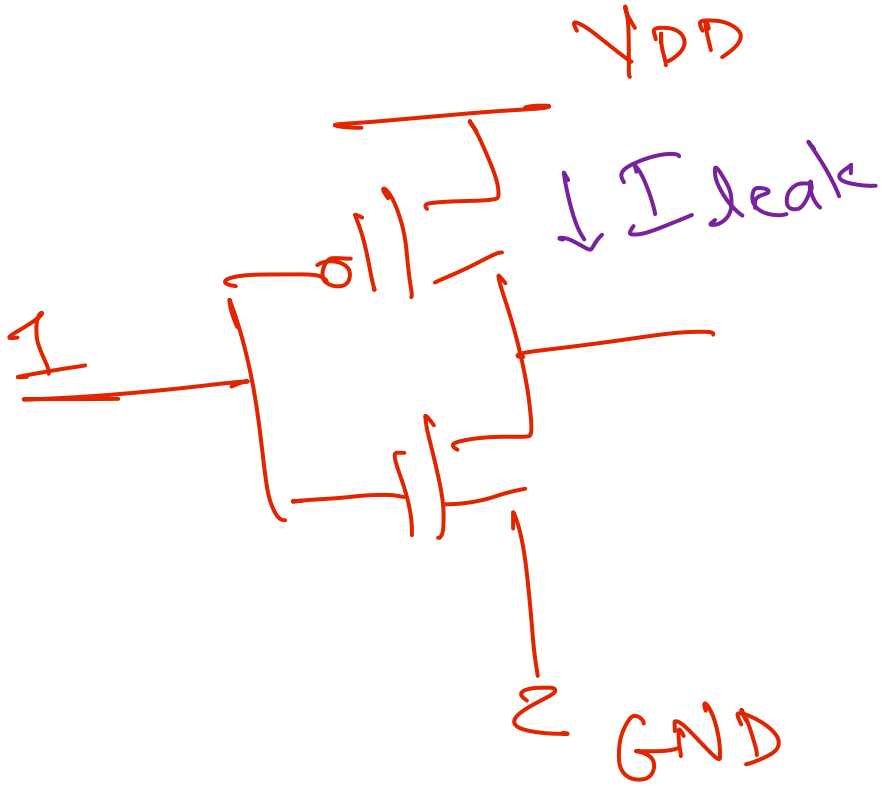
Short-circuit current



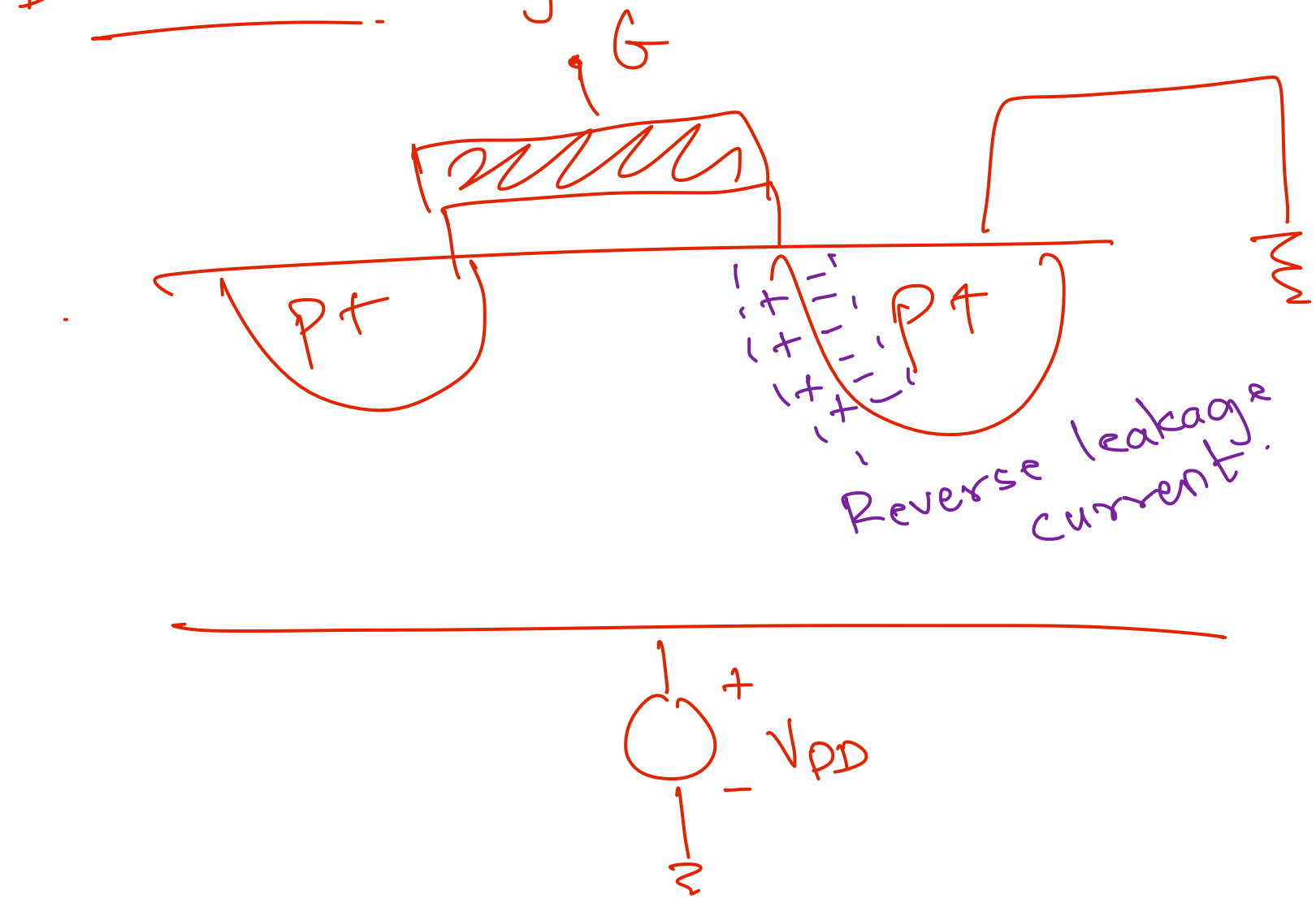
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Leakage power:

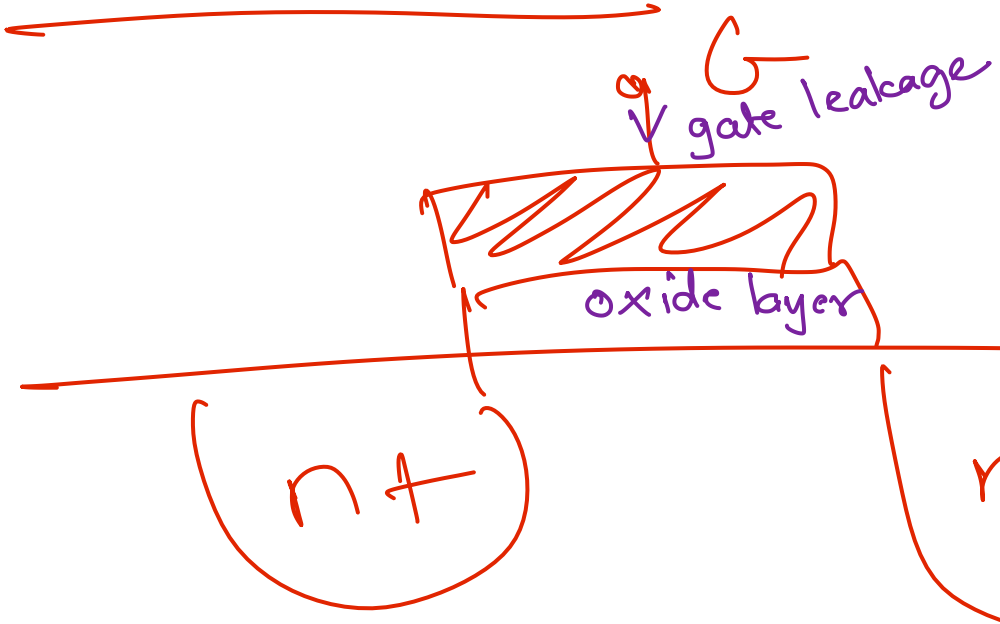


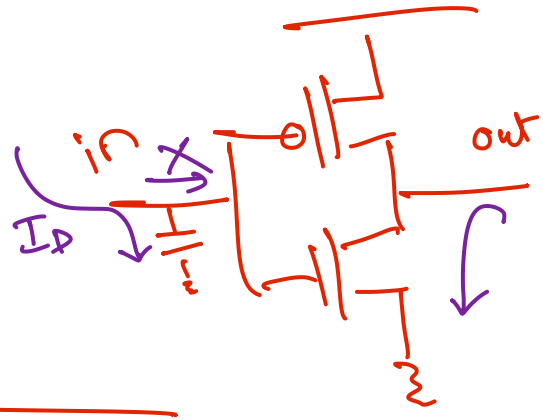


Diode leakage



Gate leakage.





Transistor leakage

$$I_D = I_0 e^{\frac{q(V_{GS} - V_{THN})}{n k T}}$$

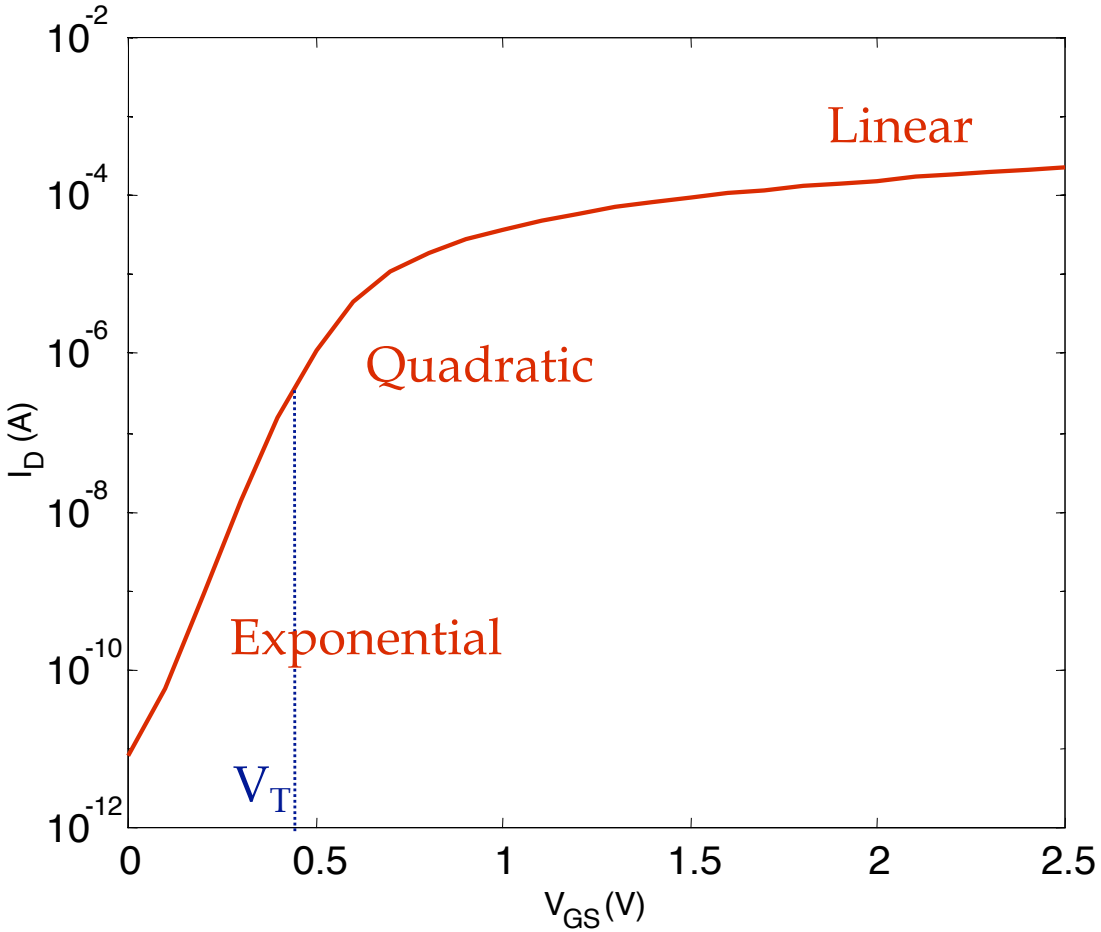
Nominal current.

$$n = 1 + \frac{C_D}{C_{ox}}$$

Depletion cap.

sub-threshold slope factor

Sub-Threshold Conduction



The Slope Factor

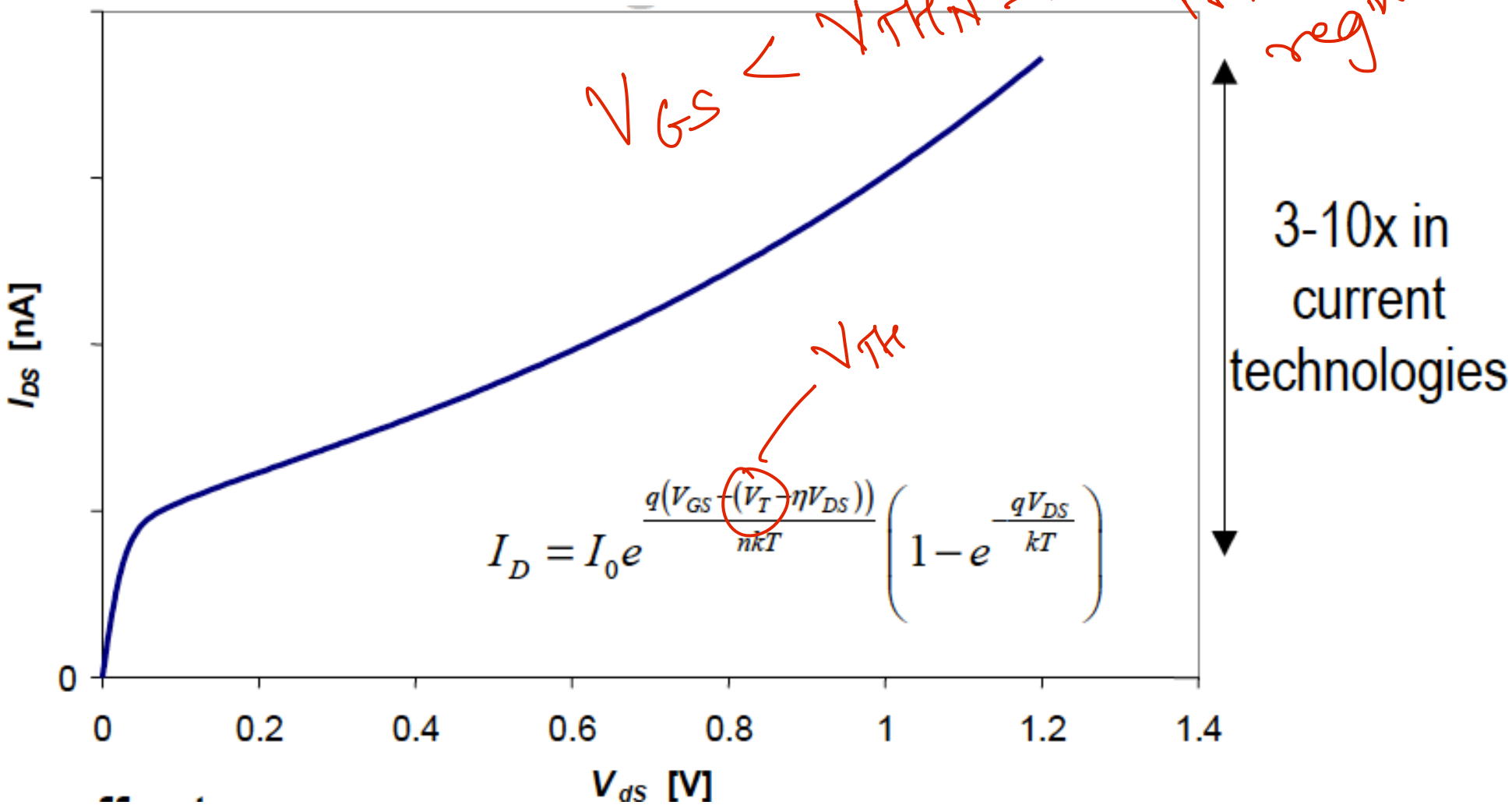
S^{-1} is ΔV_{GS} for I_{D2}/I_{D1} =10

$$S^{-1} = n\left(\frac{kT}{q}\right)\ln(10)$$

Typical values for S^{-1}:
60 .. 100 mV/decade

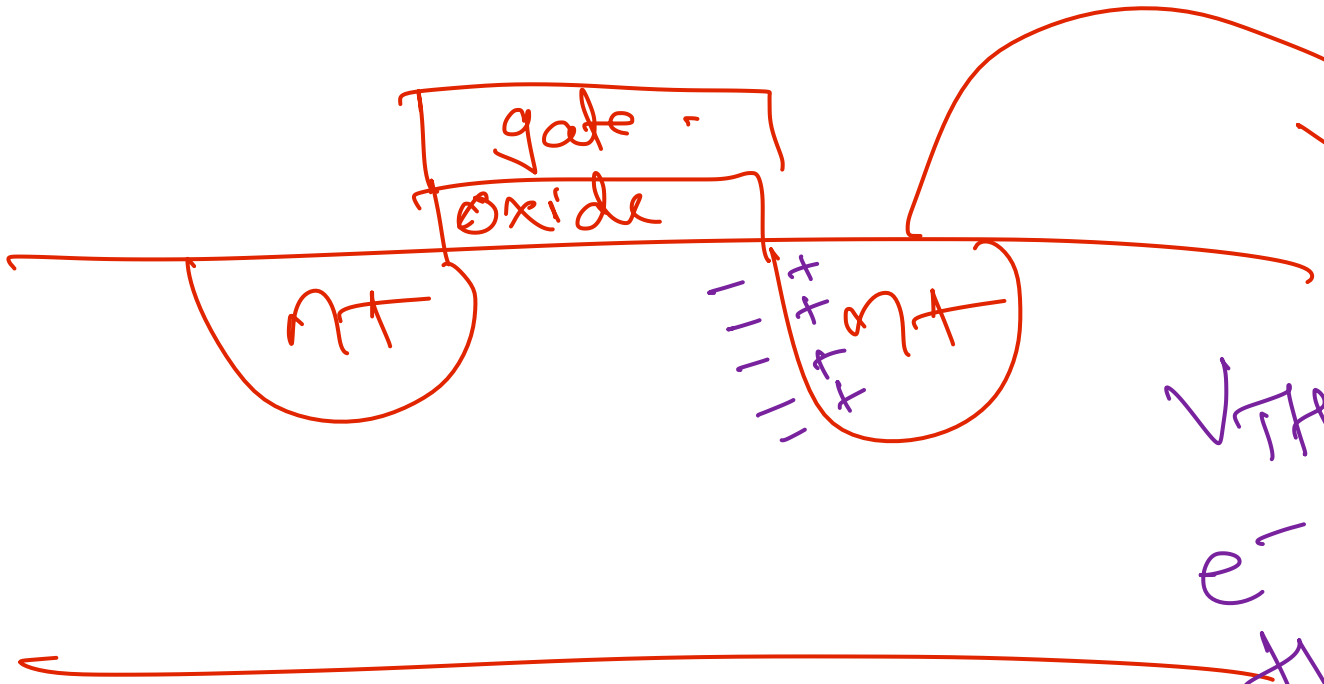
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Transistor leakage



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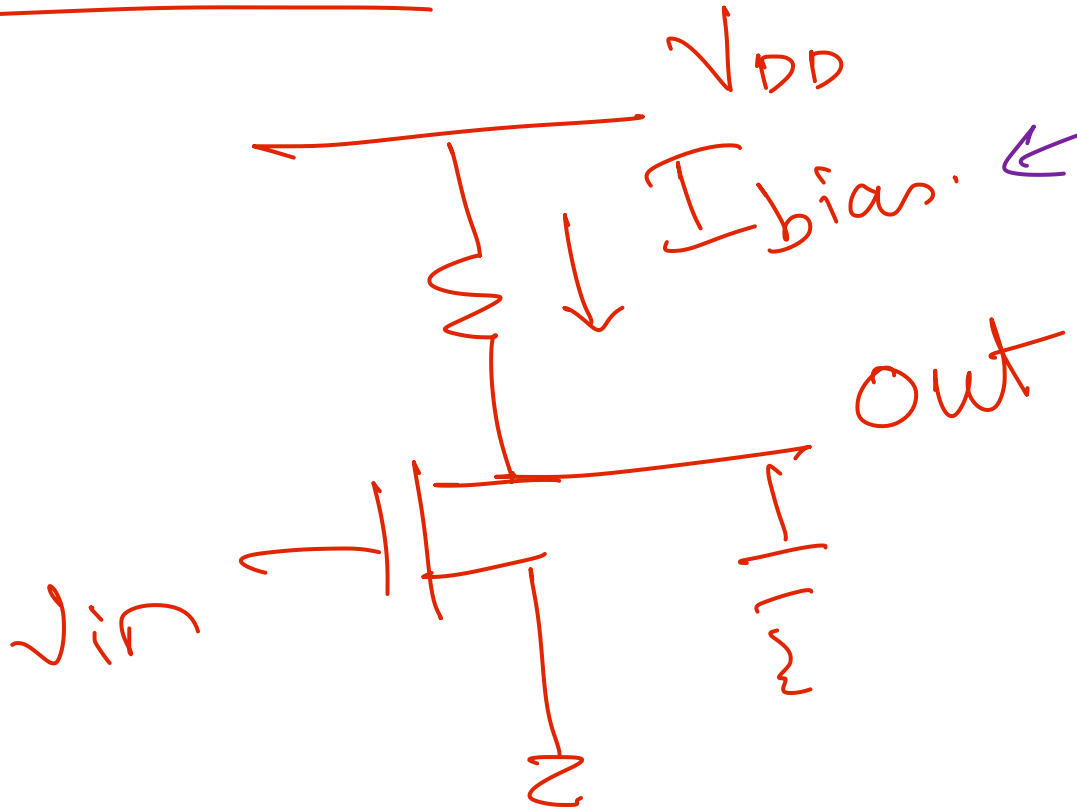
Drain Induced Barrier Lowering



V_{TH} ↓ due e^- next to the n+ region

I_{leak} ↑

Static Power



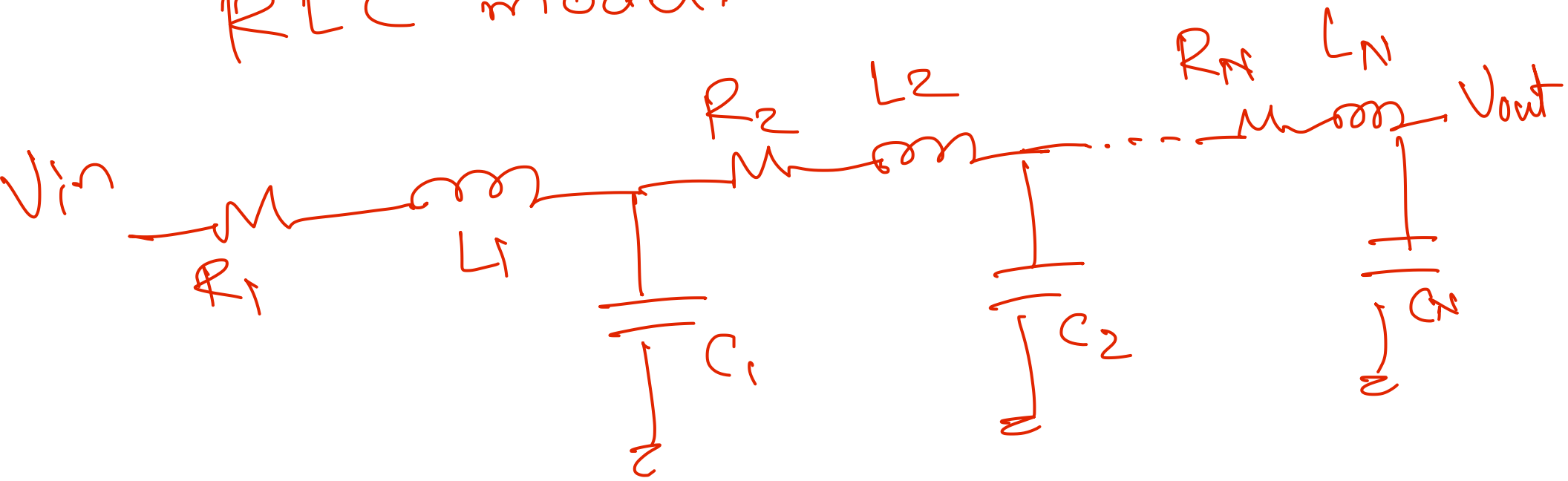
Full adder

$$S = A \oplus B \oplus C$$

$$S = SoP$$

Wires

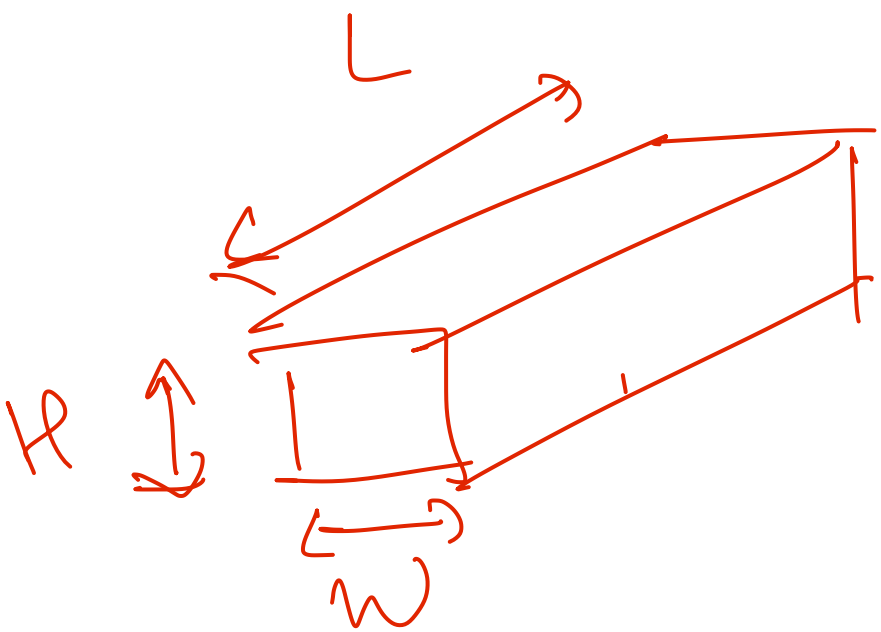
RLC model.



Resistance of a wire

$$R = \frac{\rho L}{HW}$$

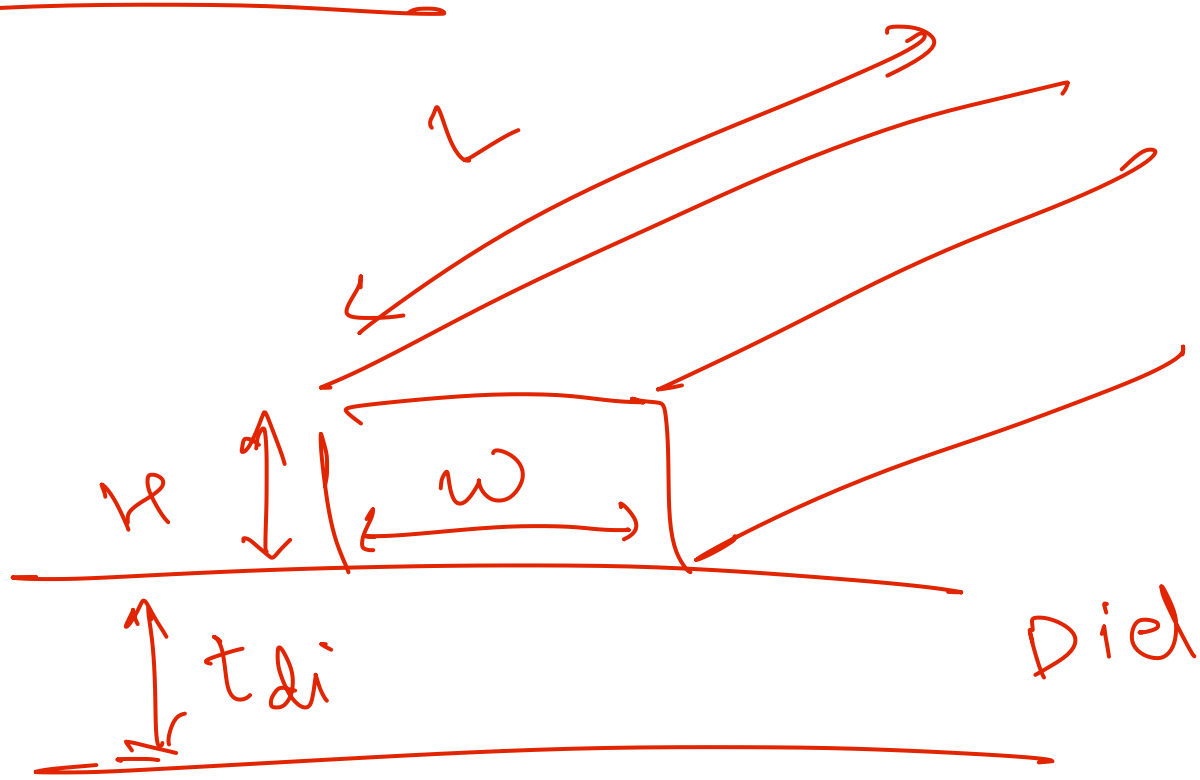
ρ — Resistivity



Cu is commonly used today

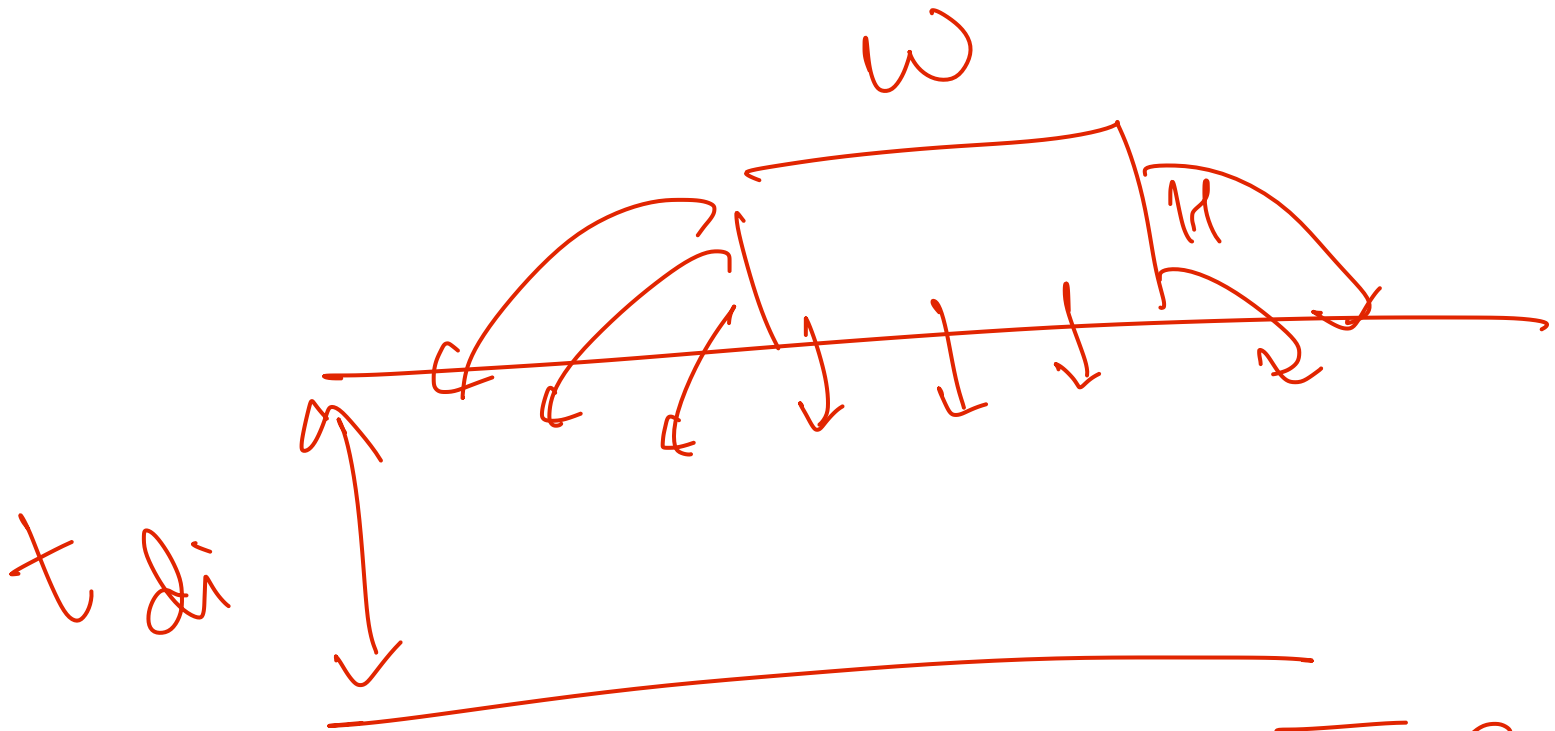
Capacitance

Parallel plate capacitance



$$C_{pp} = \frac{\varepsilon_{di} \cdot w \cdot l}{t_{di}}$$

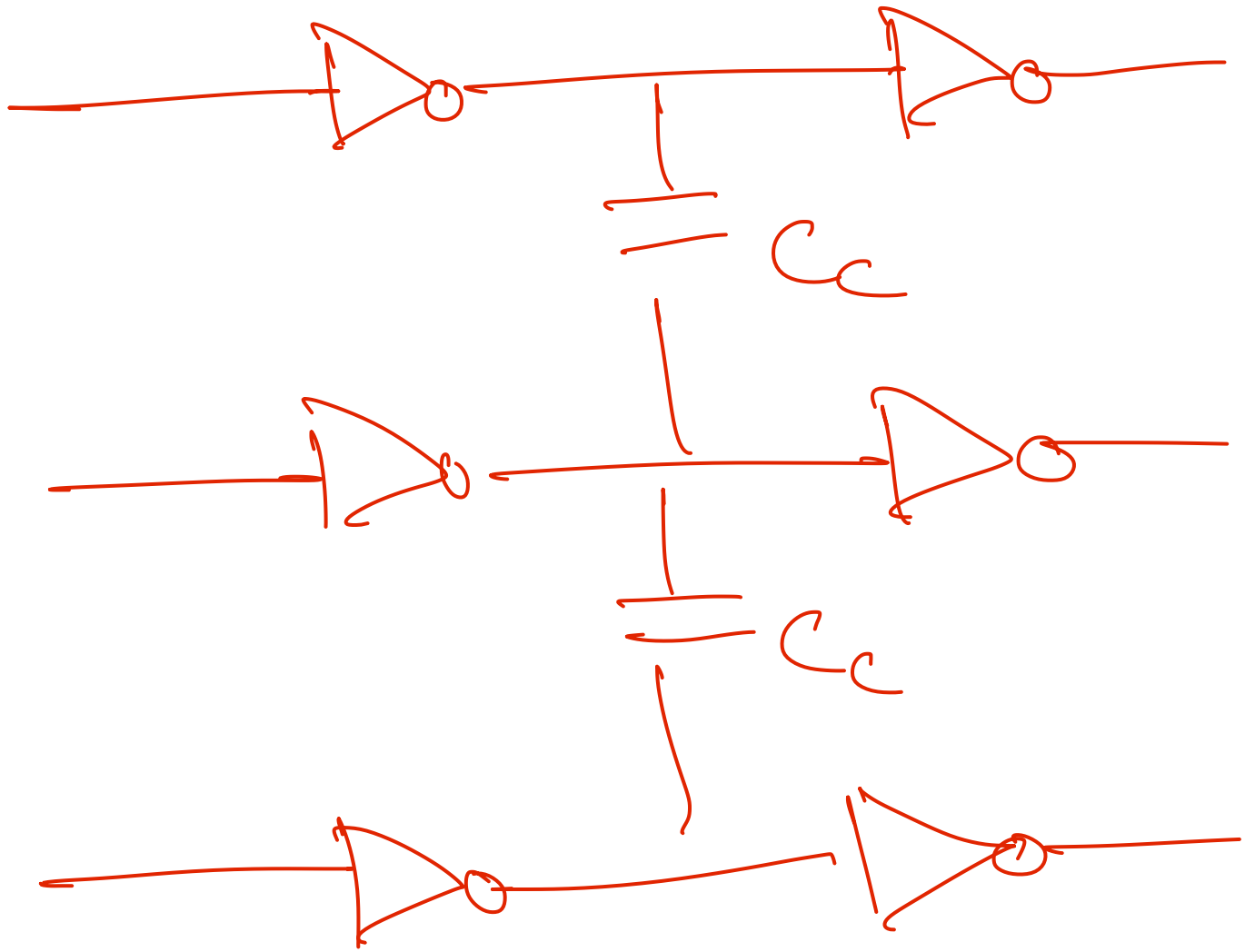
Fringing capacitance



$$C_{fringe} = \frac{2\pi \varepsilon_{di}}{\log(t_{di}/H)}$$

$$C_{wire} = C_{pp} + C_{fringe}.$$

Coupling capacitance.



(1) All wires switching in the same dirn at the same time

$C_{eff} = 0$

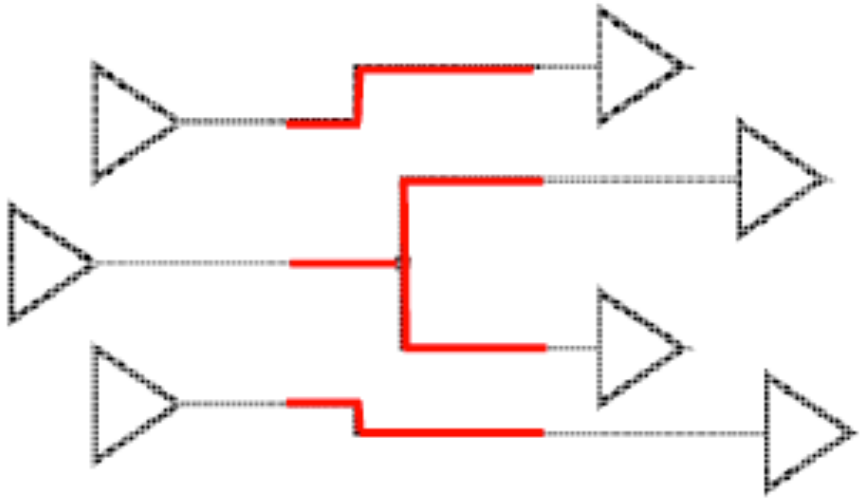
② Neighboring wires are switching in opposite direction

$$C_{eff} = 2C_c$$

③ Neighboring wires are grounded.

$$C_{eff} = C_c$$

Wires



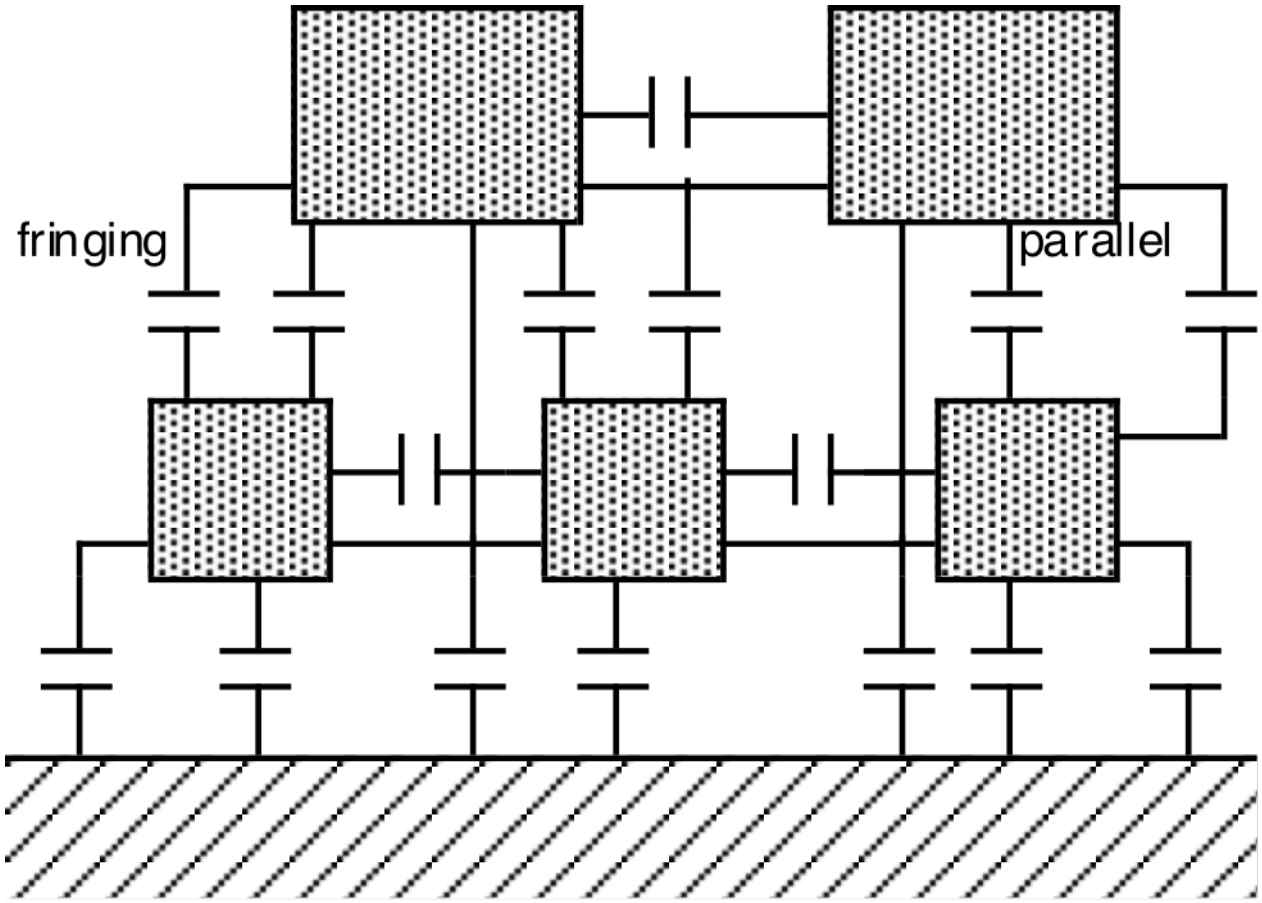
Schematic



Physical

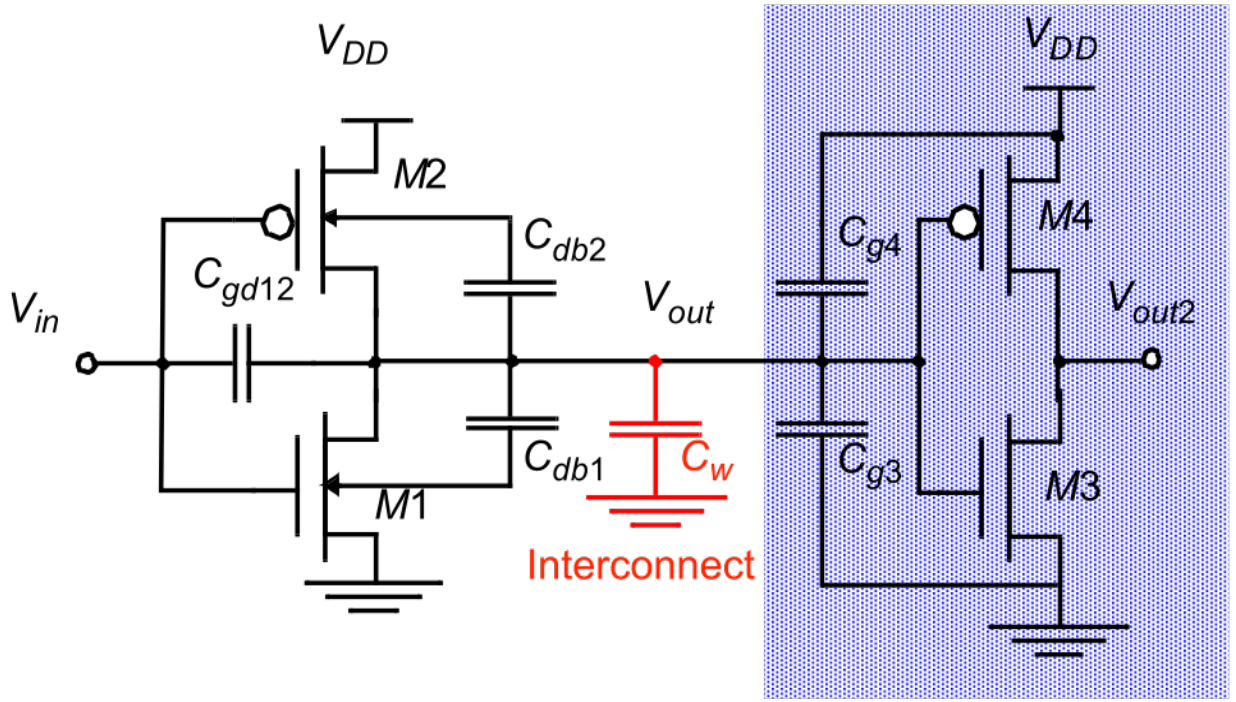
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Interwire Capacitance

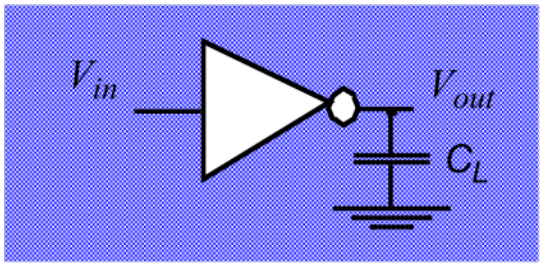


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Capacitance of Wire Interconnect



Simplified Model



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Inductance.

Design Rules of Thumb

- RC delays should only be considered when t_{pRC} >> t_{pgate} of the driving gate

$$Lcrit >> \sqrt{t_{pgate}/0.38rc}$$

- RC delays should only be considered when the rise (fall) time at the wire input is smaller than the rise (fall) time of the wire

$$t_{rise} < RC$$

  - when not met, the change in the signal is slower than the propagation delay of the wire

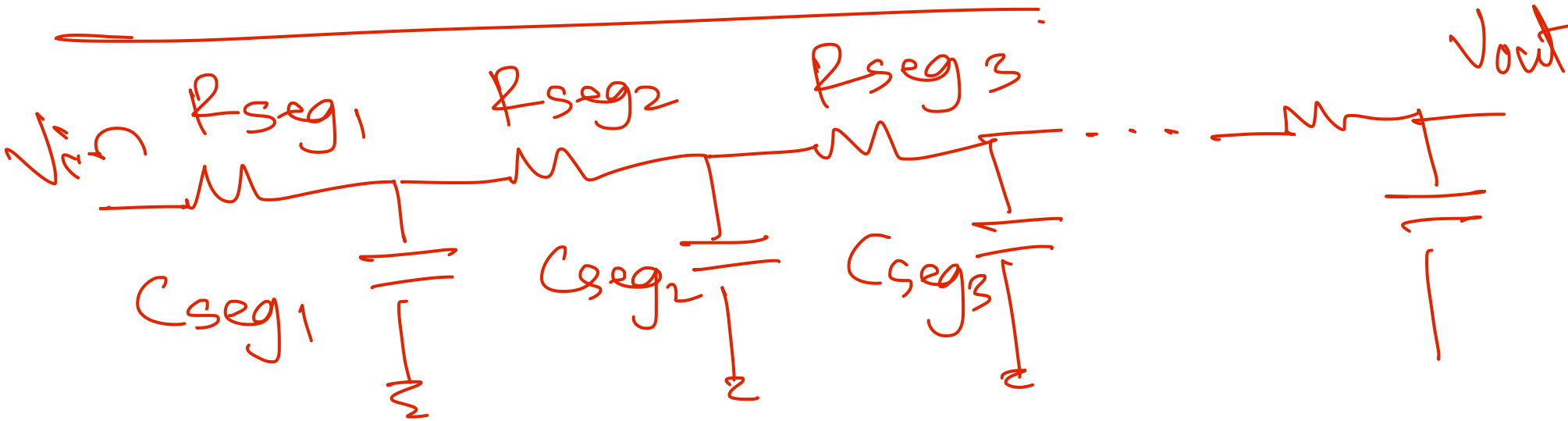
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Wire Model

Lumped model



Distributed model.



$$R_{seg1} = R_{seg2} = R_{seg3} = \cdots = r\,\Delta L$$

r: resistance / length

ΔL: segment length

$$C_{seg1} = C_{seg2} = C_{seg3} = \cdots = c\,\Delta L$$

c: capacitance / length

ΔL: segment length

$$\tau_{lumped} = R_w \cdot C_w$$

$$\tau_{distributed} = C_{seg_1}(R_{seg_1}) + C_{seg_2}(R_{seg_2} + R_{seg_1}) + C_{seg_3}(R_{seg_3} + R_{seg_2} + R_{seg_1}) + \ldots$$

$$= C_{\Delta L}(r\Delta L) + C_{\Delta L}(r\Delta L + r\Delta L) + C_{\Delta L}(r\Delta L + r\Delta L + r\Delta L) + \ldots$$

$$\Delta L = \frac{L}{N}$$

L: wire length

N: # segments

$$= \Delta L^2 rc + 2\Delta L^2 rc$$

$$+ 3\Delta L^2 rc \cdots$$

$$= rc\left(\frac{L}{N}\right)^2 (1 + 2 + 3 + \cdots)$$

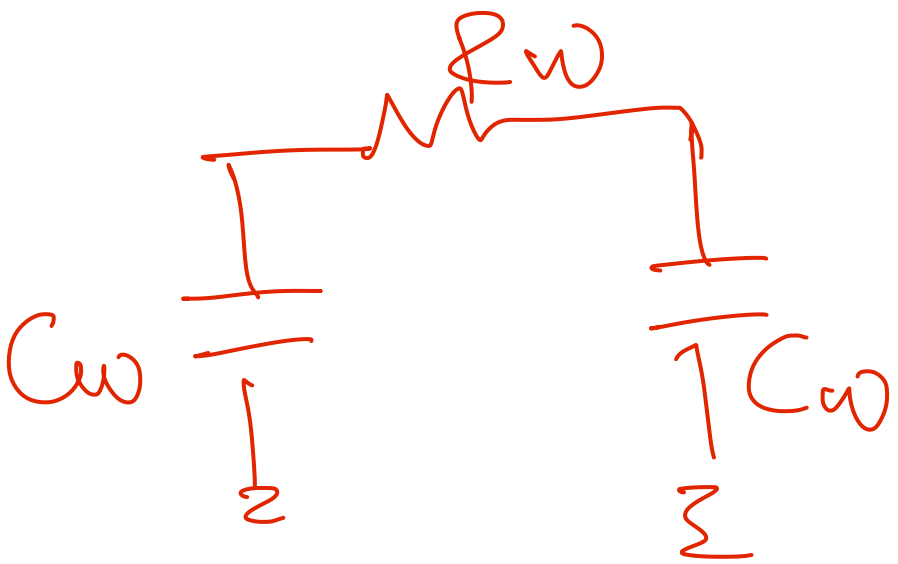
$$= \frac{rcL^2}{N^2} \cdot \frac{N(N+1)}{2}$$

If N is large

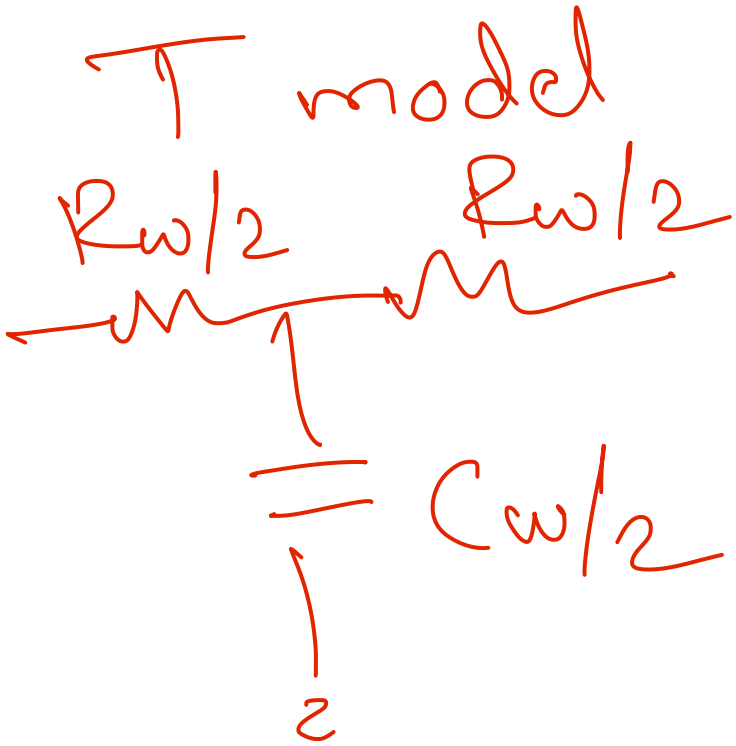
$$T_{distributed} = \frac{rcL^2}{2} = \frac{R_w C_w}{2}$$

Alternate wire models

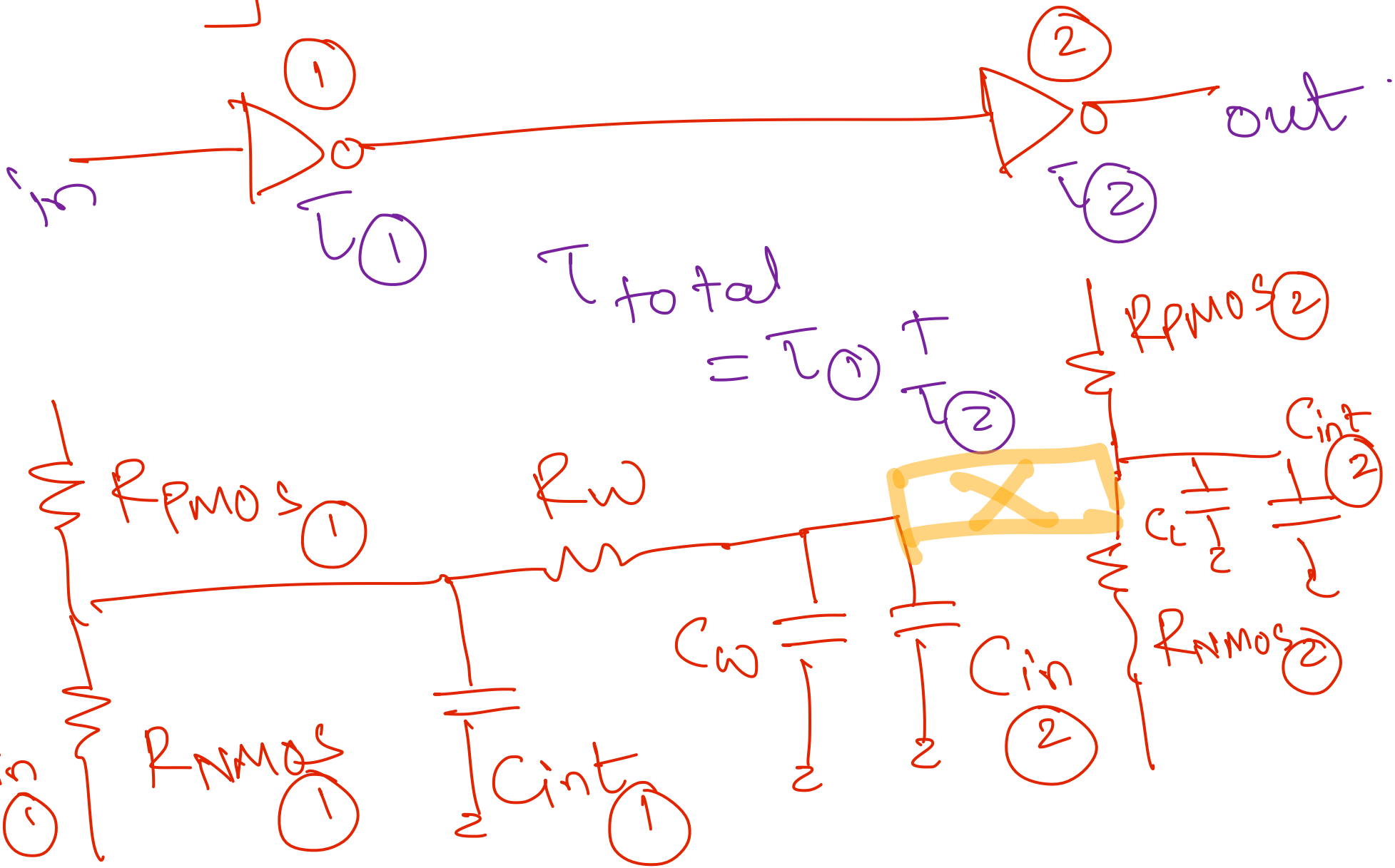
Π model.



T model



Delay model → Inverter + wire.



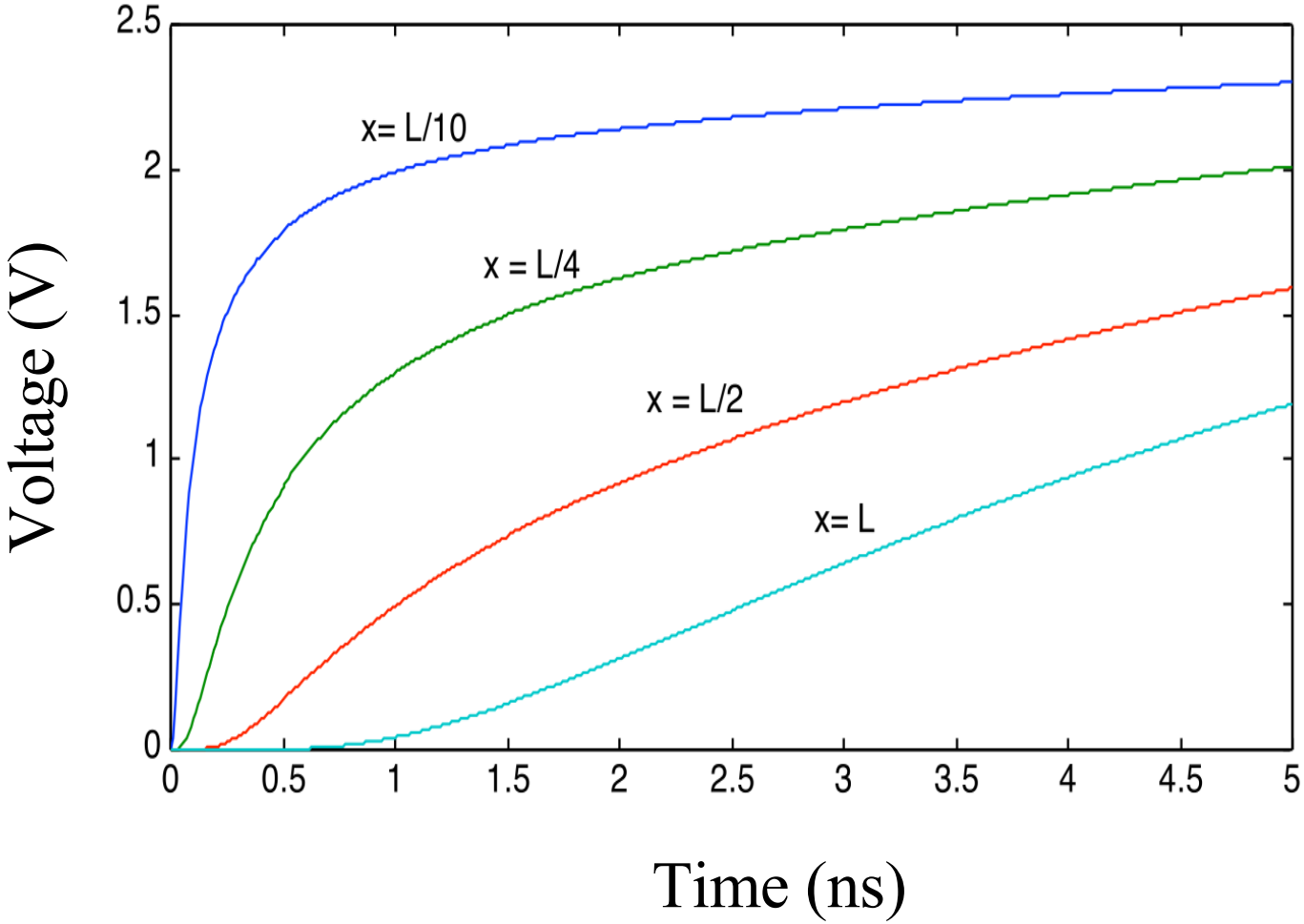
$$\tau_1 = C_{int_1}(R_{PMOS_1}) + C_w(R_w + R_{PMOS_1}) + C_{in_2}(R_w + R_{PMOS_1})$$

$$\tau_{②} = C_{int_{②}} \left(R_{NMOS_{②}}\right)$$

$$C_L \left(R_{NMOS_{②}}\right)$$

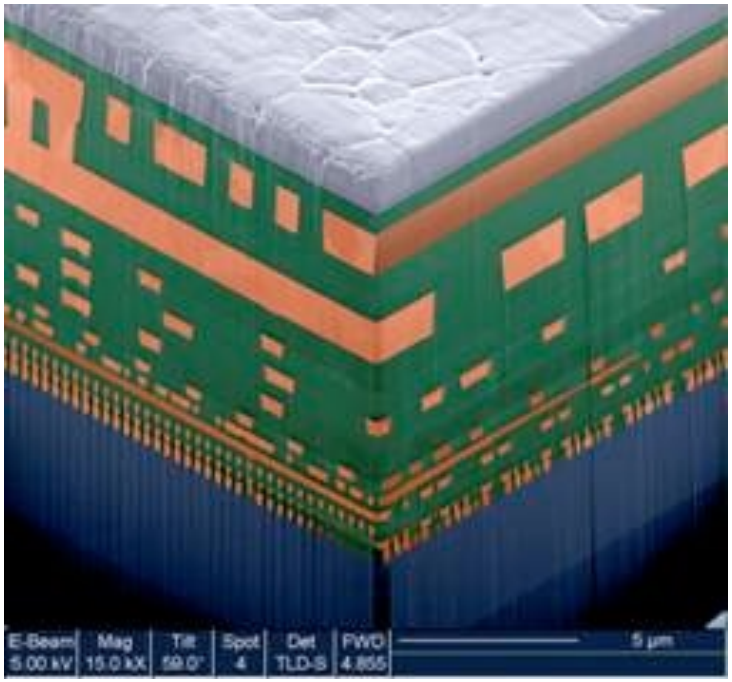
$$\tau_{total} = \tau_{①} + \tau_{②}$$

Step-response of RC wire as a function of time and space



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Wire stack in VLSI systems

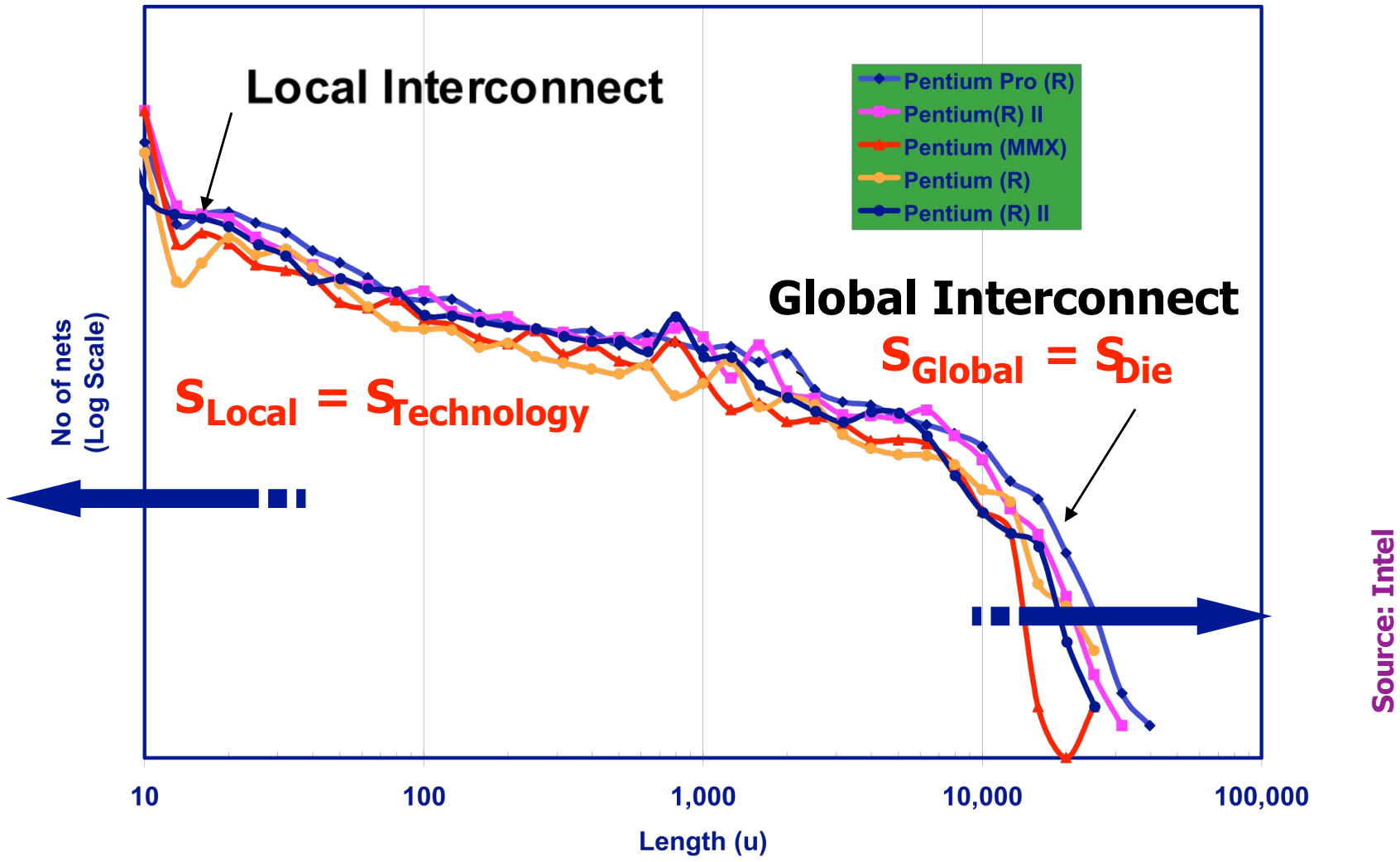


90 nm technology
Courtesy: IBM

Orthogonal routing in alternate layers to reduce coupling cap

- Wires stacked in layers above the transistors
- 10-12 metal layers in current CMOS technologies

Nature of Interconnect



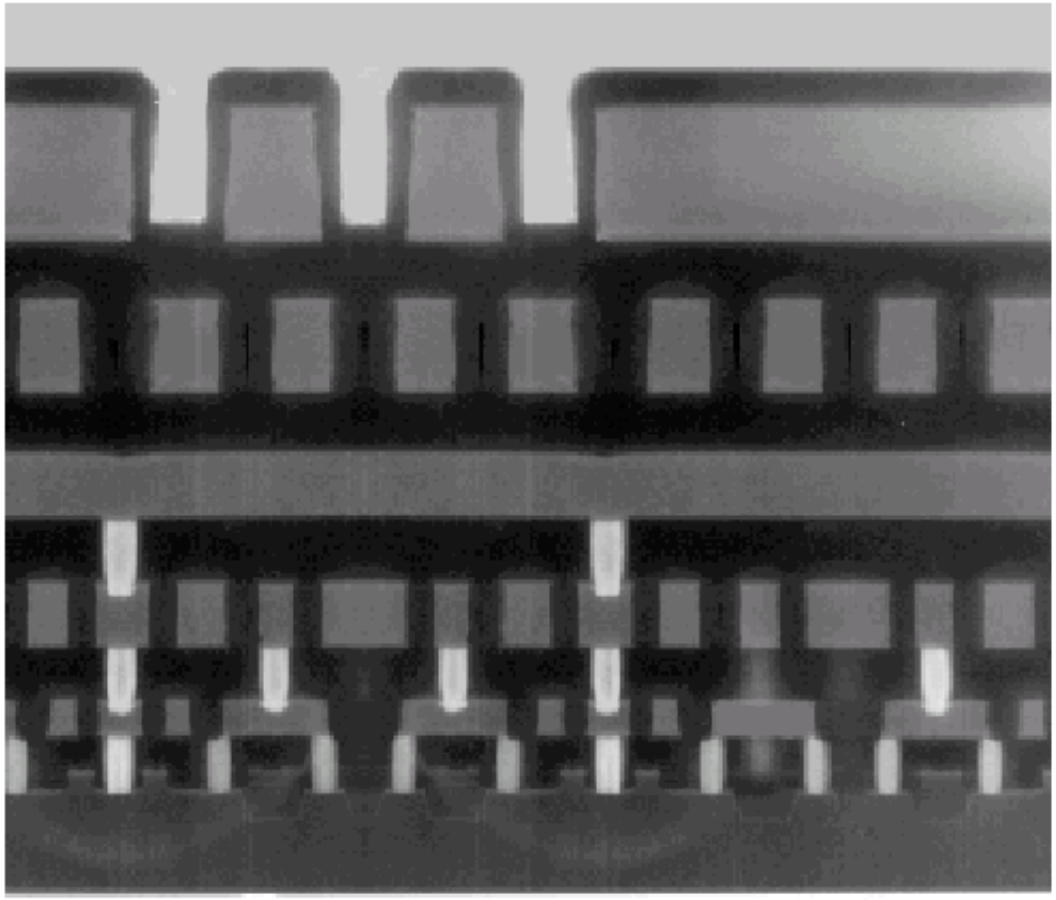
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Example: Intel 0.25 micron Process

5 metal layers
Ti/Al - Cu/Ti/TiN
Polysilicon dielectric

LAYER	PITCH	THICK	A.R.
Isolation	0.67	0.40	-
Polysilicon	0.64	0.25	-
Metal 1	0.64	0.48	1.5
Metal 2	0.93	0.90	1.9
Metal 3	0.93	0.90	1.9
Metal 4	1.60	1.33	1.7
Metal 5	2.56	1.90	1.5
	µm	µm	

Layer pitch, thickness and aspect ratio



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